

Task 22: Final Prices With a Special Discount in a Shop

Problem Description

You are given an integer array `prices` where `prices[i]` is the price of the *i*th item in a shop. There is a special discount for items in the shop. If you buy the *i*th item, then you will receive a discount equivalent to `prices[j]` where *j* is the minimum index such that *j* > *i* and `prices[j] <= prices[i]`. Return an integer array `answer` where `answer[i]` is the final price you will pay for the *i*th item of the shop.

Java Solution

```
class Solution {
    public int[] finalPrices(int[] prices) {
        int[] ans = prices.clone();
        Stack<Integer> stack = new Stack<>();

        for (int i = 0; i < prices.length; i++) {
            while (!stack.isEmpty() && prices[stack.peek()] >= prices[i])
            {
                int idx = stack.pop();
                ans[idx] -= prices[i];
            }
            stack.push(i);
        }
        return ans;
    }
}
```